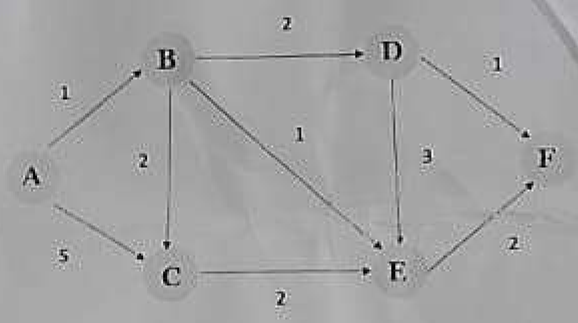
Course No: CSE 2202 || Lab Final || Total Marks: 15 || Time: 1 hours

Q1. The Department of CSE stores the marks of students from a recent exam. Each mark is an integer between 0 and 100. You are required to write a program that sorts all marks in ascending order using an efficient non-comparison-based sorting algorithm. 05

Input Format: N $m_1\ m_2\ m_3 \dots m_N$
Output Format: Sorted marks in ascending order
Constraints: • $1 \le N \le 1{,}000{,}000$ • $0 \le m_i \le 1000$ • Built-in sorting functions (like sort() or sorted()) are **not allowed**.

Q2. A passenger wants to find the shortest travel time from **City A** to all other cities in a network. Given the following weighted graph of cities and roads, compute the shortest travel time from City A to every other city. 05



Input Format: • First line: N M — number of cities and number of roads. • Next M lines: u v w — a road from city u to city v with travel time w. City A is always vertex 1.
Example: 5 7 1 2 10 1 3 5 2 3 2 3 2 3 2 4 1 3 4 9 3 5 2
Output Format: Shortest travel time from City A (City 1): City 1: 0 City 2: x City 3: y City 4: z City 5: w

Q3. A traveler is preparing for a trip and has a single bag with a weight limit. The traveler has several items to choose from, such as clothes, gadgets, and souvenirs. 05

- Each item has a weight and a value (importance or usefulness).
- The traveler cannot take a fraction of any item — each item must be taken completely or left behind.

The goal is to select a combination of items to maximize the total value carried in the bag without exceeding the weight limit.

Input Format:

$N\ W$
$w_1\ v_1$
$w_2\ v_2$
...
$w_N\ v_N$

- N = number of items
- W = maximum weight capacity of the bag
- w_i = weight of item i
- v_i = value of item i

Example:

4 7
1 1
3 4
4 5
5 7

Output Format:

Maximum value: 9

Constraints:

- $1 \leq N \leq 1000$
- $1 \leq W \leq 1000$
- $1 \leq w_i, v_i \leq 1000$